



MCQs – Software Construction

Lecture 7: AI in Software Development

(Dr. Ehab Essa Exact Exam Style)

1. Artificial Intelligence mainly enables machines to:

- A. Store data
- B. Simulate intelligence
- C. Compile code
- D. Render UI

Answer: B

2. Generative AI is best described as AI that can:

- A. Analyze logs
- B. Optimize memory
- C. Create content
- D. Encrypt data

Answer: C

3. Generative AI creates output mainly based on:

- A. Hard rules
- B. User prompts
- C. Fixed scripts
- D. Databases

Answer: B

4. Large Language Models (LLMs) are primarily used for:

- A. Image compression
- B. Text generation
- C. Database indexing
- D. Hardware control

Answer: B

5. ChatGPT became widely popular in:

- A. 2018
- B. 2020
- C. 2022
- D. 2024

Answer: C

6. AI code generation relies heavily on:

- A. SQL
- B. NLP
- C. UI design
- D. OS services

Answer: B

7. Autocompletion helps developers by:

- A. Fixing bugs
- B. Predicting code
- C. Writing tests
- D. Deploying apps

Answer: B

8. Code synthesis means:

- A. Optimizing code
- B. Writing code from description

- C. Removing bugs
- D. Formatting code

Answer: B

9. Boilerplate code refers to code that is:

- A. Complex
- B. Reusable
- C. Repetitive
- D. Obsolete

Answer: C

10. Which tool is used for AI-assisted code generation?

- A. Eclipse
- B. GitHub Copilot
- C. Jenkins
- D. Maven

Answer: B

11. AI bug detection is based mainly on:

- A. Syntax rules
- B. Code patterns
- C. User feedback
- D. Compiler errors

Answer: B

12. Error prediction aims to:

- A. Fix UI
- B. Predict future bugs
- C. Improve UX
- D. Refactor code

Answer: B

13. Automated debugging allows AI to:

- A. Rewrite requirements
- B. Suggest fixes
- C. Deploy software
- D. Manage teams

Answer: B

14. AI testing automation mainly reduces:

- A. Accuracy
- B. Coverage
- C. Manual effort
- D. Security

Answer: C

15. AI-generated test cases are often based on:

- A. Compiler output
- B. User stories
- C. Database schema
- D. Network logs

Answer: B

16. Test optimization focuses on:

- A. Running all tests
- B. Prioritizing tests
- C. Removing tests
- D. Manual testing

Answer: B

17. Auto-documentation uses AI to:

- A. Optimize code
- B. Write comments
- C. Generate documentation
- D. Compile programs

Answer: C

18. AI documentation tools mainly use:

- A. SQL
- B. NLP
- C. JVM
- D. APIs

Answer: B

19. Refactoring with AI improves:

- A. Code readability
- B. Hardware speed
- C. UI colors
- D. Network latency

Answer: A

20. Performance optimization aims to:

- A. Change behavior
- B. Reduce memory/time
- C. Add features
- D. Improve UI

Answer: B

21. AI security tools help detect:

- A. Syntax errors
- B. SQL injection
- C. Memory allocation
- D. UI bugs

Answer: B

22. Code auditing mainly focuses on:

- A. UI design
- B. Security flaws
- C. Speed
- D. Styling

Answer: B

23. Generative AI affects SDLC by:

- A. Removing developers
- B. Automating stages
- C. Eliminating testing
- D. Increasing cost

Answer: B

24. Requirement extraction uses AI to:

- A. Write code
- B. Read documents
- C. Compile software
- D. Deploy apps

Answer: B

25. Ambiguous requirements are handled by AI through:

- A. Ignoring them
- B. Clarification
- C. Deletion
- D. Compilation

Answer: B

26. Design and planning AI tools generate:

- A. Bytecode
- B. UML diagrams
- C. Logs
- D. Databases

Answer: B

27. Prompt Driven Development (PDD) treats the prompt as:

- A. Comment
- B. Specification
- C. Test case
- D. Output

Answer: B

28. Prompt chaining is based on which strategy?

- A. Trial-and-error
- B. Divide-and-conquer
- C. Random search
- D. Exhaustive search

Answer: B

29. One risk of a single large prompt is:

- A. Faster output
- B. Instruction neglect

- C. Better clarity
- D. Lower cost

Answer: B

30. Good prompting requires improving:

- A. Memory and speed
- B. Context, specificity, scope
- C. Hardware and OS
- D. UI and UX

Answer: B